WEEK 1 · 9–12 NOVEMBER 2026

Spectral signatures and sensor concepts

Why materials absorb where they do, and what the sensor does to that signal

Week 1 · Session 3 · practical: 10_spectral_signatures_and_libraries.ipynb

LEARNING OBJECTIVES

What you will be able to do

- Explain the two physical mechanisms that produce absorption features in the 400–2500 nm range
- Identify diagnostic features of vegetation, water, soils, minerals and burnt surfaces by wavelength
- Use continuum removal and band depth to compare features between bright and dark materials
- Define band centre, FWHM, spectral sampling, SNR and ground sampling distance, and say why each matters
- Judge whether two materials remain separable once a spectrum is resampled to a multispectral sensor

1. Why materials have spectra

Electrons and molecular bonds

Two mechanisms, one spectrum

- **Electronic transitions** — an electron absorbs a photon and jumps to a higher energy level
  - Strong, broad features; dominate the visible and near-infrared (400–1000 nm)
- **Vibrational transitions** — molecular bonds stretch and bend at characteristic frequencies
  - Fundamental vibrations lie in the mid-infrared; what we see in the SWIR are their *overtones* and *combination bands* — weaker, but narrow and diagnostic
- Everything between 400 and 2500 nm is one of these two, plus scattering

Electronic transitions: colour and iron

- **Chlorophyll a and b** absorb strongly near 430 and 460 nm (blue) and 640 and 680 nm (red)
  - The gap between them is why healthy vegetation looks green
- **Iron** dominates soil and rock colour: Fe^{3+} crystal-field absorption near 850–900 nm
  - Charge-transfer absorption in the blue and ultraviolet makes iron-rich soils red
- Other pigments: carotenoids (~450–500 nm), anthocyanins (~530–550 nm), phycocyanin (~620 nm)
- These features are **broad** — a multispectral sensor can often see them

Vibrational overtones: what the SWIR reveals

- **O–H in liquid water** — strong at ~1450 and ~1940 nm, weaker at ~970 and ~1200 nm
- **Al–OH in clays** — ~2200 nm; kaolinite shows a characteristic doublet near 2160 and 2206 nm
- **Mg–OH** — ~2300–2350 nm, in chlorites and serpentines
- **CO_3 in carbonates** — ~2340 nm, with a secondary feature near 2160 nm
- **C–H in cellulose, lignin and hydrocarbons** — ~1700, ~2100 and ~2300 nm
- **Gypsum** — hydrated sulphate, with features near 1450, 1750 and 1940 nm
  - Methane absorbs near 2300 nm — the same window, which is why it competes with clay features

HOW TO READ A SPECTRUM

A spectrum is an inventory of bonds and electrons.

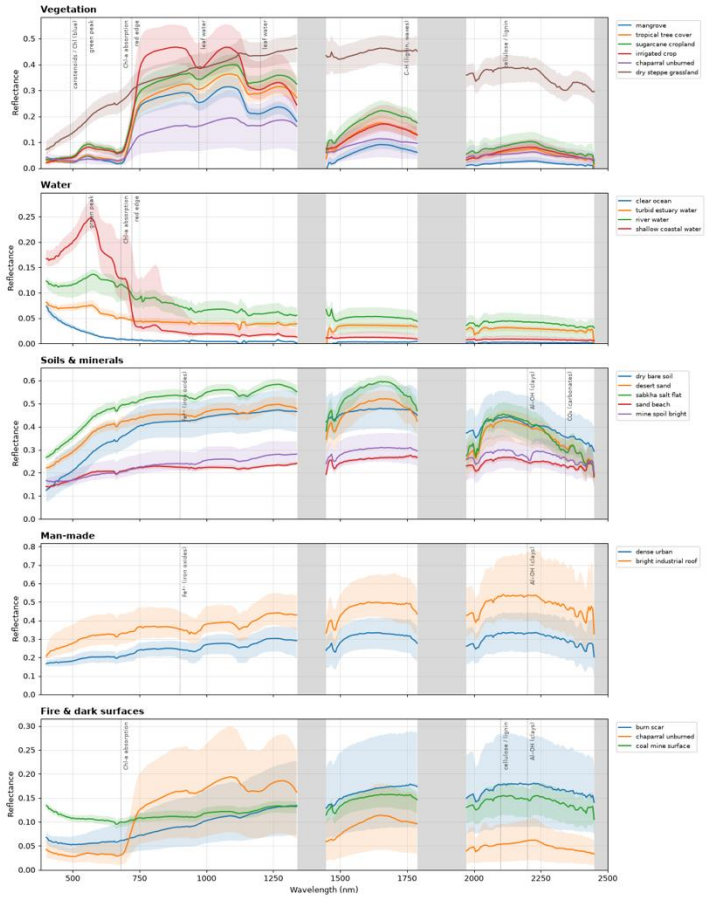
Read it that way and the features stop being curve shapes to memorise: each dip names a molecule or a mineral that is present in the pixel.

2. The diagnostic features

Wavelength by wavelength, material by material

MEASURED, NOT TEXTBOOK

The course reference library



19 image-derived spectra from the seven course scenes, grouped by material family

Diagnostic features to memorise

Wavelength	Feature	Tells you about
430 / 460, 640 / 680 nm	Chlorophyll absorption	Photosynthetic activity, pigment content
690–750 nm	Red edge	Chlorophyll concentration, stress, species
850–900 nm	Fe^{3+} crystal field	Iron oxides in soils and spoil
970, 1200 nm	Weak liquid water	Leaf and canopy water content
1450, 1940 nm	Strong liquid water	Moisture (also atmospheric water vapour)
~1700, 2100 nm	C–H, cellulose and lignin	Dry plant matter, crop residue
2160 / 2206 nm	Al–OH doublet	Clay mineralogy (kaolinite, illite, smectite)
~2340 nm	CO_3 and Mg–OH	Carbonates, altered minerals

Vegetation: the most structured spectrum in nature

- Low reflectance in blue and red (pigment absorption), a small green peak near 550 nm
- **Red edge** — the steep rise between about 690 and 750 nm
  - Its inflection point shifts to **longer** wavelengths as chlorophyll increases (Guyot & Baret 1988 give a four-point linear estimate)
- NIR plateau (750–1300 nm): multiple scattering at cell wall–air interfaces, set by leaf structure
- SWIR troughs at 1450 and 1940 nm deepen as canopy water increases
- As a leaf dries, water features weaken and **cellulose and lignin near 2100 nm** emerge

Water: darkness with structure

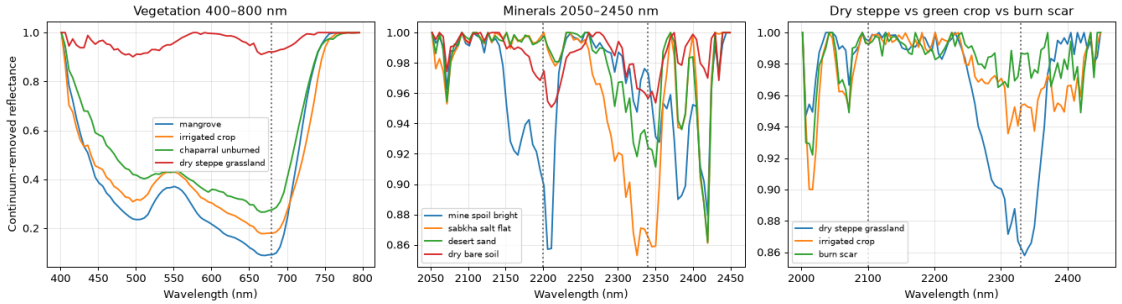
- Water absorbs increasingly towards the NIR — beyond ~900 nm, clear water is essentially black
- **All the information is in the visible**, where light penetrates and returns
- **Chlorophyll-a** in phytoplankton: absorption near 440 and 665 nm, fluorescence peak near 685 nm
- **Phycocyanin** near 620 nm marks cyanobacteria — a public-health signal
- **Suspended sediment** raises reflectance across the visible and shifts the peak towards red
- **Shallow water** adds bottom reflectance — sand, seagrass and coral mix into the signal

Soils, minerals and man-made surfaces

- Bare soil rises smoothly from blue to SWIR; the **slope** is set by iron and organic matter
- Superimposed on that slope: narrow Al–OH, Mg–OH and CO_3 features that name the mineralogy
- **Evaporites** — sabkha, salt flats, gypsum — are bright with hydration features
- **Burnt surfaces**: char and ash collapse the NIR plateau and raise SWIR reflectance
  - This inversion is what NBR measures, and why burn scars are unmistakable in the SWIR
- **Man-made materials** — concrete, asphalt, metal and plastic roofing — have their own signatures

Continuum removal: comparing features fairly

- Divide the spectrum by its upper convex hull
- Brightness differences cancel; only the **feature shape** remains
- **Band depth** = 1 − R(centre) / R(continuum)
- Now a dark coal surface and a bright spoil heap can be compared directly



Raw spectra (left) and their continuum-removed form (right)

REAL NUMBERS FROM NOTEBOOK 10

Band depths measured in the course library

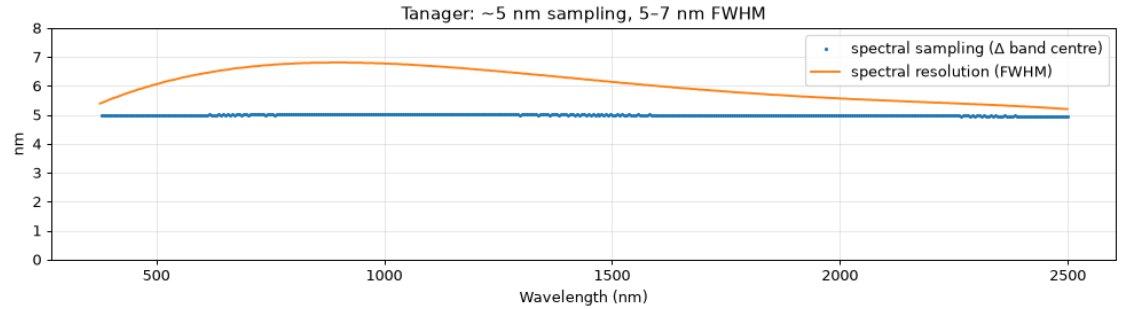
Material	Chl 680	Al–OH 2200	CO_3 2340
Mangrove	0.891	−0.204	0.070
Irrigated crop	0.784	−0.150	0.015
Chaparral, unburned	0.669	−0.120	0.019
Mine spoil, bright	0.030	0.094	−0.001
Sand beach	0.024	0.044	0.024
Sabkha salt flat	−0.027	−0.027	0.094

3. What the sensor does to the signal

Band centre, FWHM, SNR, pixel size

Band centre, sampling and resolution

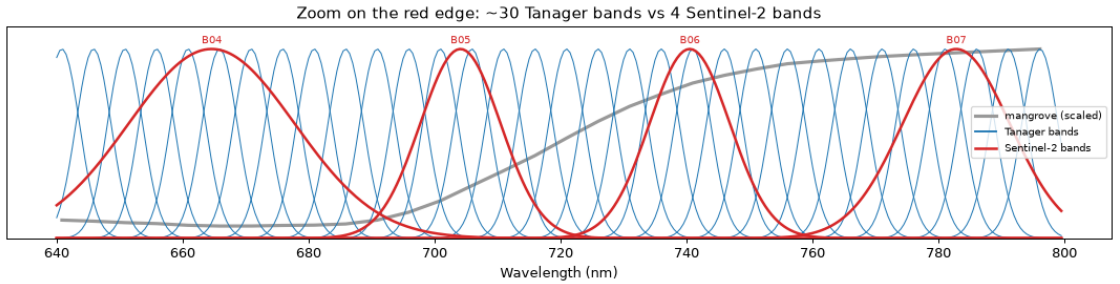
- **Band centre** — the wavelength a band is tuned to
- **Spectral sampling** — the spacing between adjacent centres (Tanager ≈ 5 nm)
- **Spectral resolution** — the width of one band's response, its **FWHM** (≈ 5.2–6.8 nm)
- Sampling and resolution are different numbers, and both matter



Tanager band centres and their widths

The same scene through four sensors

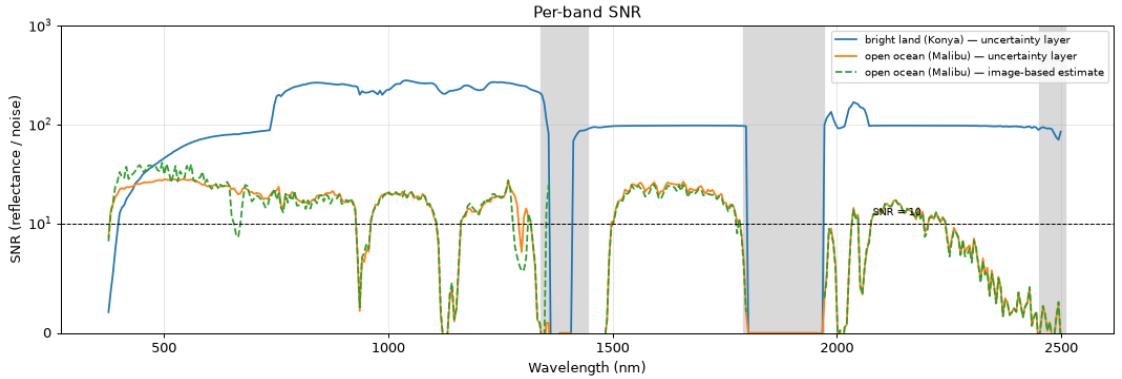
- Each curve is a band's spectral response function
- Tanager: 426 bands · EnMAP: 224 · Sentinel-2: 13 · Landsat 8/9: 7 reflective
- Sentinel-2 B12 spans about 175 nm — the whole Al–OH and CO_3 region in one number
- **A feature narrower than a band cannot be recovered afterwards**



Spectral response functions on a common axis

Signal-to-noise: not all bands are equal

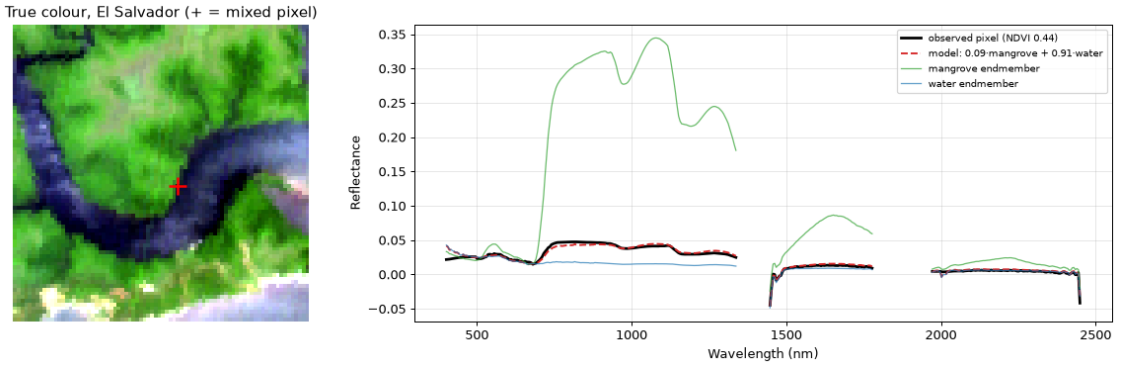
- SNR is highest where solar irradiance and surface reflectance are high
- It collapses below ~400 nm: little solar energy, weak detector response
- It collapses inside the water-vapour windows: the photons never reach the ground
- It falls at the SWIR detector edge beyond ~2450 nm
- **Consequence:** we drop those bands before analysis, as notebook 11 shows



Per-band SNR estimated from the Tanager uncertainty layer

One pixel, several materials

- At 30 m, most pixels mix several materials
- Mixing is approximately **linear in area fraction** for side-by-side materials
- A measured pixel of NDVI 0.44 turned out to be **9 % mangrove, 91 % water**
- Indices mislead on mixed pixels; **unmixing** estimates the fractions instead



A mixed pixel decomposed into its endmembers

The trade-offs you cannot escape

- Splitting light into 426 narrow bands leaves fewer photons per band → larger pixels preserve SNR
- **Ground sampling distance:** Tanager and EnMAP both sit at 30 m for this reason
- **Swath:** narrow (tens of kilometres), so acquisitions are **tasked**, not continuous
- **Revisit:** irregular — the Tanager open archive holds 153 unique scenes so far
- **Plan projects around what was acquired**, not around what you wish had been

4. Libraries, resampling and separability

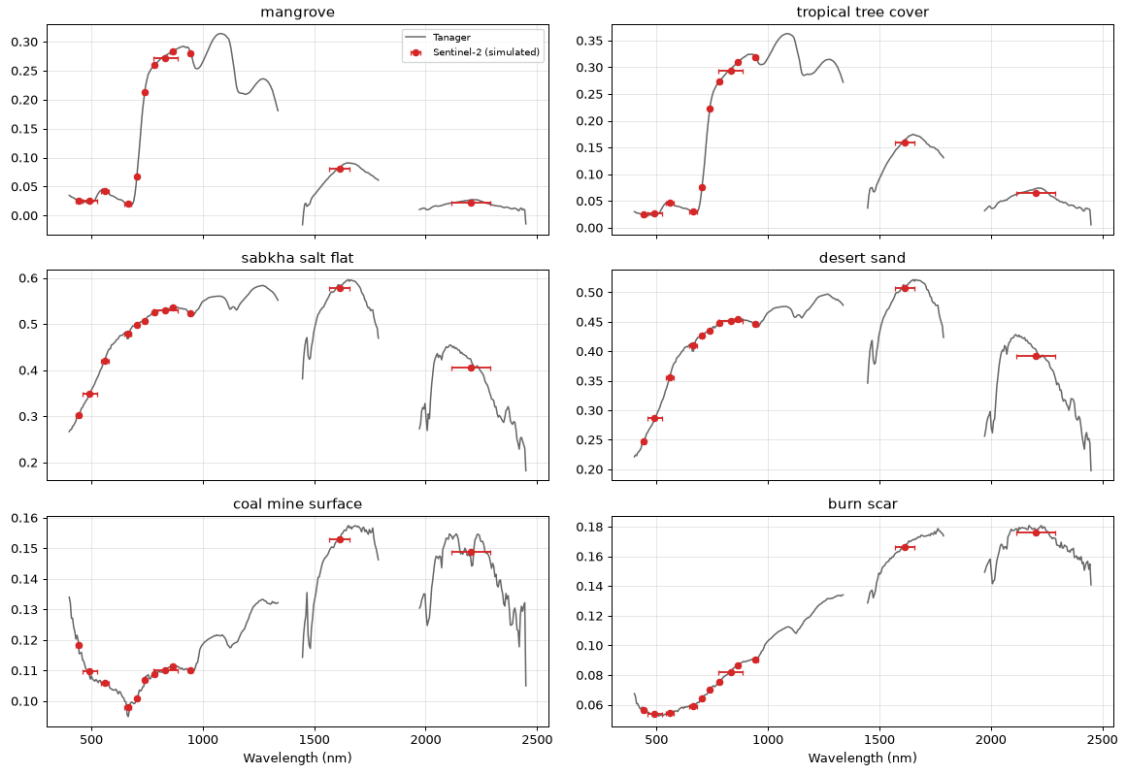
Can you still tell the materials apart?

Two kinds of spectral library

- **Laboratory libraries** — USGS splib07, ECOSTRESS: pure samples, controlled illumination
  - Ideal reference shapes, but no atmosphere, no mixing, no canopy structure
- **Image-derived libraries** — mean spectra of homogeneous areas in your own scene
  - Same sensor, same atmosphere, same mixing — directly comparable to your pixels
- The course ships an image-derived library of 19 materials, built from pure WorldCover pixels
- Use lab spectra to *identify*, image spectra to *classify*

What Sentinel-2 would see

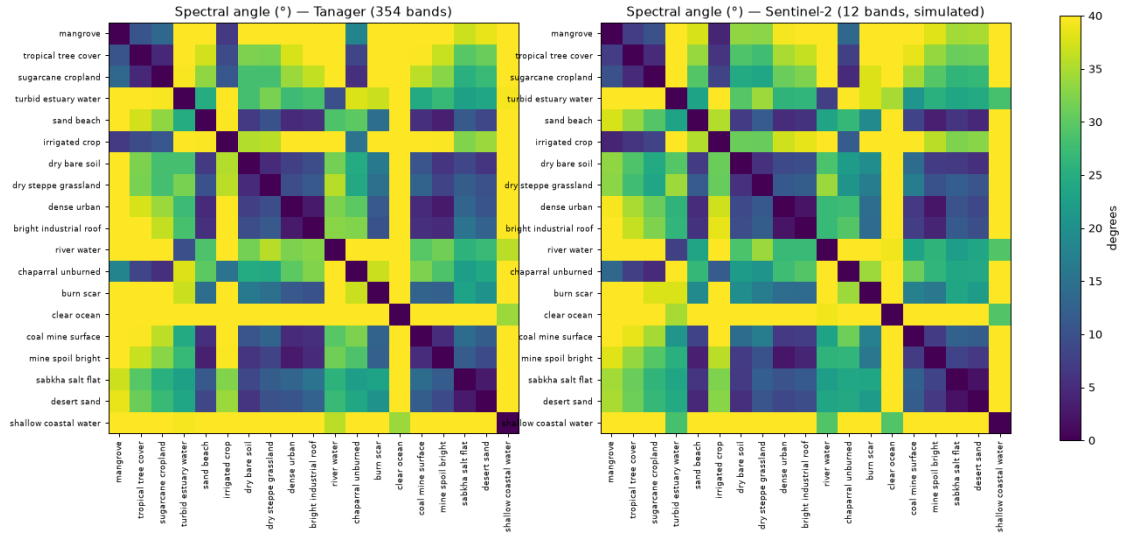
- Convolve each library spectrum with Sentinel-2's response functions
- The broad shape survives: vegetation still looks like vegetation
- The **narrow features vanish** — the clay doublet becomes a single number
- This is the honest test of whether your project needs hyperspectral data



Library spectra before and after resampling to Sentinel-2

Separability: the spectral angle

- The angle between two spectra treated as vectors
- Insensitive to brightness — it compares **shape**, not magnitude
- Small angle = hard to tell apart; large angle = easily separable
- Across all 171 library pairs, **71 % became less separable** after resampling to Sentinel-2



Pairwise spectral angles within the course library

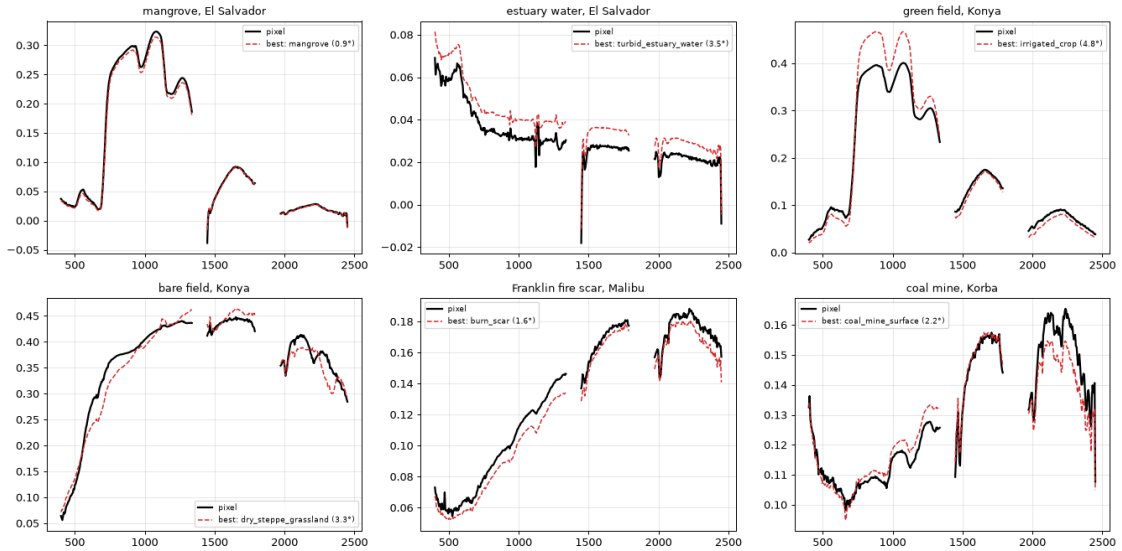
MEASURED IN NOTEBOOK 10

Which pairs lose the most

Material pair	Tanager	Sentinel-2	Retained
Mangrove vs tropical tree cover	10.44°	7.24°	69 %
Sabkha vs desert sand	2.91°	1.97°	68 %
Irrigated crop vs sugar cane	10.85°	8.23°	76 %
Dense urban vs bright mine spoil	2.83°	2.49°	88 %
Coal mine surface vs burn scar	12.69°	17.35°	137 %

Matching a real pixel to the library

- Pick a pixel, compute the angle to every library spectrum, take the smallest
- Mangrove pixel matched *mangrove* at **0.9°**; the burn scar matched *burn_scar* at 1.6°
- The second-best match tells you how confident the identification is
- This is Spectral Angle Mapper — the basis of the week 2 detection session



Nearest library match for six hand-picked pixels

So when do you actually need a spectrum?

- **Yes** — identifying materials: minerals, roofing, crop residue, algal pigments
- **Yes** — detecting gases: the methane window at 2300 nm
- **Yes** — sub-pixel work: unmixing fractions rather than mapping a dominant class
- **Maybe** — stress detection: narrow red-edge bands beat broad ones, but not always decisively
- **No** — mapping obvious extent: vegetation, water, built-up; Sentinel-2 is cheaper and more frequent
  - Say so in your project when multispectral would do — the rubric rewards honesty

PRACTICAL

Over to you

▶ `10_spectral_signatures_and_libraries.ipynb` · `90 min`

- Plot the library by material family and find the features from the cheat-sheet table
- Apply continuum removal and measure band depths for clay, carbonate and chlorophyll
- Compare Tanager, EnMAP, Sentinel-2 and Landsat response functions and SNR
- Resample the library to Sentinel-2 and compute how much separability is lost
- Match real pixels to the library with the spectral angle

KEY TAKEAWAYS

Recap

- Features come from **electronic transitions** (VNIR) and **vibrational overtones** (SWIR)
- Learn the wavelengths: 680 chlorophyll · red edge · 970/1200/1450/1940 water · 2100 cellulose · 2200 Al–OH · 2340 CO_3
- **Continuum removal and band depth** make features comparable between bright and dark materials
- Sampling, FWHM, SNR and GSD decide which questions the data can answer
- Resampling to Sentinel-2 cost separability in **71 % of library pairs** — that is the value of narrow bands

Next: Session 4 — Hyperspectral data access, visualisation and exploration

References and further reading

Clark, R.N. & Roush, T.L. (1984). Reflectance spectroscopy: quantitative analysis. JGR 89(B7), 6329–6340.

Clark, R.N. (1999). Spectroscopy of rocks and minerals, and principles of spectroscopy. Manual of Remote Sensing, Vol. 3.

Guyot, G. & Baret, F. (1988). Utilisation de la haute résolution spectrale pour suivre l'état des couverts végétaux.

Kruse, F.A. et al. (1993). The Spectral Image Processing System (SIPS) — the Spectral Angle Mapper. RSE 44, 145–163.

Kokaly, R.F. et al. (2017). USGS Spectral Library Version 7. USGS Data Series 1035.

Meerdink, S.K. et al. (2019). The ECOSTRESS spectral library version 1.0. RSE 230, 111196.